
Question

In base-9, what is $67+25$? Please reason it step by step. End the response with the result in "Answer: result".

0-shot (Correct)

To add in base-9, we start by adding the digits in the rightmost place value and moving leftwards. In the units place, $7+5=12$. However, in base-9, there are only digits from 0 to 8. Therefore, we need to do a "carry" to the next place value. The digit 12 is written as 13 in base-9, since the digit 12 in base-9 is represented as 10, followed by 11. Now, we move to the next place value, the 9s place. We have a carry of 1 from the units place, so the addition becomes $1+6+2=9$. In base-9 notation, 9 is written as 10. Therefore, the sum of $67+25$ in base-9 is 103. Answer: 103.

1-shot (Wrong)

In base-9, the digits are "012345678". We have $7 + 5 = 12$ in base-10. Since we're in base-9, that doesn't exceed the maximum value of 8 for a single digit. $12 \bmod 9 = 3$, so the digit is 3 and the carry is 1. We have $1 + 6 + 2 = 9$ in base-10. $9 \bmod 9 = 0$, so the digit is 0 and the carry is 1. We have $1 + 1 + 0 = 2$ in base-10. So the final digit is 2. Therefore, the answer is 203. Answer: 203.

3-shot (Correct)

In base-9, the digits are "012345678". We have $7 + 5 = 12$ in base-10. Since we're in base-9, that exceeds the maximum value of 8 for a single digit. $12 \bmod 9 = 3$, so the digit is 3 and the carry is 1. We have $6 + 2 + 1 = 9$ in base 10. $9 \bmod 9 = 0$, so the digit is 0 and the carry is 1. A leading digit is 1. So the answer is 103. Answer: 103.

Table 44: Case for (CWC): 0-shot correct, 1-shot wrong, 3-shot correct.

Task	Setting	#Method		
		Base Model	Variant of CD-CoT	CD-CoT
Base-9	irr.-med	30.3	47.0	60.7
	ina.-med	10.1	47.0	58.7
Sym.(E)	irr.-easy	28.1	33.0	44.7
	ina.-easy	29.1	32.0	42.6

Table 45: Performances of different variants of CD-CoT.

On the NoRa-Math Base-9 dataset, the LLM answers incorrectly in the initial 20 attempts. When we increase N to 50, the LLM gets it right twice. We use one of the self-generated correct rationales to execute CD-CoT, testing the effect of LLM’s self-generated rationale. On the NoRa-Symbolic Equal dataset, we need to increase N to 100 to get a correct self-generated response.

Regarding the empirical results in Tab. 45, we derive the following detailed analysis.

Observation F.14 (The self-supervised variant of CD-CoT achieves enhancements across various tasks). On NoRa-Math Base-9, the average accuracy is increased by 26.8 percentage points. On NoRa-Symbolic Equal, it gains an average increase of 3.9 points in accuracy.

Observation F.15 (LLM struggles to generate a clean and correct rationale on its own). Considering the fact that the model requires 50 attempts on the simple NoRa-Math Base-9 task to obtain the correct result, it indicates that the LLM struggles to generate a clean and correct rationale on its own. Additionally, experiments on the NoRa-Symbolic Equal indicate that as the task’s difficulty increases, it becomes more challenging for the model to resolve it by relying solely on its inherent capabilities.

Observation F.16 (Self-generated clean examples are not as effective as manually designed clean examples). Although the self-supervised variant of CD-CoT acquires improvement across tasks, its gains on the difficult NoRa-Symbolic Equal tasks are quite modest, with an average increase of only 3.9 points. Conversely, the original CD-CoT registers an average accuracy enhancement of 39.5 points on NoRa-Math Base-9 tasks and 15.1 points on NoRa-Symbolic Equal tasks, respectively. We can conclude that self-generated clean examples can provide some guidance to the LLM on simpler tasks, but they are not as effective as manually designed clean examples. On more difficult tasks, the guidance provided by self-generated clean examples is very limited, and these tasks require additional supervision to steer the model effectively.

In addition to the self-supervised CD-CoT, exploring strategies like using contrasting noisy examples or incorporating an external knowledge base, possibly through a retrieval-augmented denoising framework, may offer significant advances in automation and robustness of reasoning. We believe that these are promising directions that deserve further investigation.

F.8 New Datasets and Large-scale Real Problems

Recall that the constructed NoRa benchmark covers five prevailing datasets from three different domains. The current LLMs present significant vulnerability to noisy rationales in all five datasets of NoRa, while the proposed method CD-CoT has shown advanced and consistent robustness against noisy rationales. Specifically,

- Robustness with different datasets: CD-CoT consistently outperforms other methods in all five datasets in NoRa.
- Robustness with different noise levels: The results shown in Tab. 8 demonstrate the remarkable robustness of CD-CoT to varying noise levels. Across the NoRa-Math, NoRa-Symbolic, and NoRa-Commonsense tasks, the performance decline of CD-CoT remains modest as the noise level increases.
- Robust with different LLMs: The results in Tab. 11 further indicate that CD-CoT substantially improves over all three other baselines on the more powerful LLMs. Even on the relatively smaller Mistral-8x7B, CD-CoT significantly outperforms the other baselines on most tasks.

Note that Sec. 3 introduces a general framework for generating noisy rationales with existing datasets. This means, more datasets can be integrated into NoRa if necessary in future research. Here, we conduct additional experiments to generalize CD-CoT to three new datasets that are not covered in NoRa, including GSM-8K [12], Blocksworld [82], and BIG-Bench Hard Dyck Languages [72]. Specifically,

- GSM-8K: A math dataset of linguistically diverse grade school math word problems.
- Blocksworld: A planning dataset simulating block stacking tasks.
- BIG-Bench Hard Dyck Languages: A symbolic dataset designed for predicting the sequence of closing parentheses in a Dyck-4 word.

Then, we generate noisy rationales and compare the following setups:

- Zero-shot: Base model with no demonstration.
- CoT (clean rationales): Base model with 3 clean demonstrations.
- CoT (noisy rationales): Base model with 3 noisy demonstrations.
- CD-CoT (noisy rationales): Base model with 3 noisy demonstrations and our CD-CoT method.

The reasoning accuracy in Tab. 46 shows that CD-CoT consistently outperforms the zero-shot setting and standard CoT prompting under noisy rationales. This is consistent with the findings in Sec. 5.2, showing CD-CoT’s strong capability of generalization to new datasets.

Dataset	Setup	#Method			
		Zero-shot	CoT (clean)	CoT (noisy)	CD-CoT (noisy)
GSM-8K	300 samples	84.3	87.7	84.3	86.0
Blocksworld	200 samples	2.0	25.0	13.0	25.5
Hard Dyck Languages	250 samples	12.4	40.8	29.2	35.2

Table 46: Performances of different methods on new datasets.

Along with the introduction of new datasets, we also examine the issue of noisy rationale in large-scale real-world scenarios, particularly in conversational question-answering (QA) tasks. While our primary focus is on reasoning tasks, we discuss the differences and relationships between these areas.

Conversational QA tasks are usually conducted in a zero-shot setting, where users directly input the task instructions and hardly provide examples to the model. NoRa, on the other hand, considers the in-context learning scenario, where the model is given a few examples before answering questions. In-context learning can significantly enhance the model’s performance on complex or new tasks. The purpose of NoRa is to assess the model’s robustness to noisy examples in in-context learning.

However, for multi-turn conversational QA tasks, the content of previous answers can be considered as context for the model when answering the next question. Therefore, by evaluating how noisy context affects a model’s subsequent question responses, we can explore the issue of noisy rationales within the realm of conversational QA tasks.

We conduct experiments on MT-Bench [108] to investigate the noisy rationale problem in QA scenarios. MT-Bench is a multi-turn question collection and covers eight principal types of tasks: Writing, Roleplay, Extraction, Reasoning, Math, Coding, Knowledge I, and Knowledge II.

For clean context, we directly use the original question to prompt the model, and the first round of dialogue generated serves as the clean context. For noisy context, we prompt the model to insert some irrelevant information as noise when answering the first turn’s question, and the first round of dialogue serves as the noisy context. We then continue to use the second turn’s question along with the generated context to prompt the model to obtain the target response.

We conduct conversational QA tasks on GPT-3.5-turbo-0613 and score the generated responses with GPT-4. The GPT-4 is to score the responses from both the first and second turns separately.

Setting	#Avg.score	
	First Response	Second Response
Clean Context	8.73	8.23
Noisy Context	8.16	7.75

Table 47: Average scores with different contexts.

Round	Setting	#Task							
		Writing	Roleplay	Extraction	Reasoning	Math	Coding	STEM	Humanities
First Response	Clean Context	9.8	8.45	9.3	6.9	8.9	7.0	9.7	9.8
	Noisy Context	9.75	8.0	8.5	7.7	7.5	6.7	8.6	8.5
Second Response	Clean Context	9.5	9.3	8.9	5.2	5.8	7.85	9.4	9.9
	Noisy Context	9.3	9.1	8.5	4.6	4.1	7.2	9.2	10.0

Table 48: Average scores across types of tasks with different contexts.

As the results shown in Tab. 47, the noisy context decreases both the scores (quality) of the first-round and second-round QA. Specifically, the average scores across different types of tasks are shown in Tab. 48. Notably, a 0.48% decrease in average score appears in the second response, which is the final target of the conversation.

In addition, we provide a case study for illustration. As shown in Tab. 49, the noisy context in round 1 leads to a bad, noisy response in round 2.

In the clean context, the model provides an accurate analysis and answer to the initial question. When the correct round serves as the context for the second question, the model consistently delivers a precise analysis and response.

In the noisy context, the model’s response and analysis of the initial question contain errors. Specifically, as shown in Tab. 49, it incorrectly counts multiple integers for both case 1 and case 2, culminating in a wrong answer for the first question. When the flawed round serves as the context for the second question, the model generates the same intermediate results as in the first round and ultimately provides an incorrect answer to the second question.

In summary, during multi-turn conversational QA tasks, the model’s response to the current question is influenced by the preceding dialogue content (*i.e.*, the context), particularly for tasks that require reasoning and mathematical computations.